$$\int_{-\infty}^{1} 0\,dx + \int_{1}^{4} Cx^2\,dx + \int_{4}^{\infty} 0\,dx = 1.$$

$$0 + C\left[\frac{x^3}{3}\right]_1^4 + 0 = 1, \quad \Rightarrow C\left[\frac{64-1}{3}\right] = 1 \quad \Rightarrow 21C = 1 \quad \Rightarrow C = \frac{1}{21}.$$

Therefore the probability density function is

$$f(x) = \begin{cases} \frac{1}{21}x^2 & 1 < x < 4 \\ 0 & \text{Otherwise} \end{cases}$$

Since $f(x)$ is continuous, the probability that X is equal to any particular value is zero. Therefore when the random variable is continuous, either or both of the signs < by ≤ and > by ≥ can be interchanged. Thus

(i) $P(1.5 < X < 3.5) = P(1.5 \le X < 3.5) = P(1.5 < X \le 3.5) = P(1.5 \le X \le 3.5)$

Therefore

$$P(1.5 < X < 3.5) = \int_{1.5}^{3.5} f(x)\,dx = \frac{1}{21}\int_{1.5}^{3.5} x^2\,dx$$

$$= \frac{1}{21}\left(\frac{x^3}{3}\right) = \frac{1}{21}\left(\frac{(3.5)^3 - (1.5)^3}{3}\right)$$

$$= \frac{79}{126}.$$

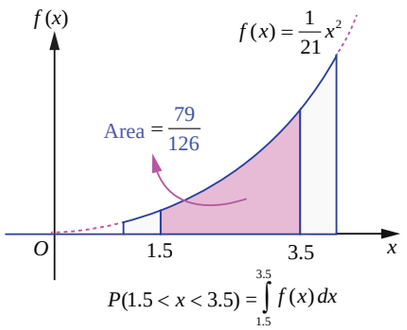


Fig. 11.13

(ii) $P(X \le 2) = \int_{-\infty}^{2} f(x)\,dx = \int_{-\infty}^{1} f(x)\,dx + \int_{1}^{2} f(x)\,dx$

Therefore

$$P(X \le 2) := 0 + \frac{1}{21}\int_1^2 x^2\,dx = \frac{1}{21}\left(\frac{x^3}{3}\right)_1^2$$

$$= \frac{1}{21}\left(\frac{2^3 - 1^3}{3}\right) = \frac{7}{63}.$$

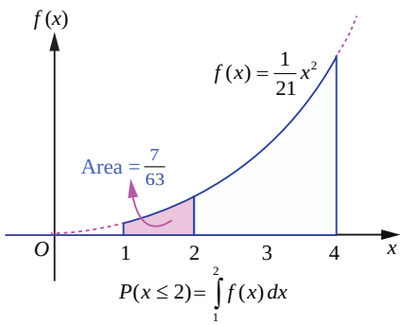


Fig. 11.14

(iii) $P(3 < X) = \int_{3}^{\infty} f(x)\,dx = \int_{3}^{4} f(x)\,dx + \int_{4}^{\infty} f(x)\,dx$

$$= \frac{1}{21}\int_3^4 x^2\,dx + 0 = \frac{1}{21}\left(\frac{x^3}{3}\right)_3^4.$$

$$= \frac{1}{21}\left(\frac{4^3 - 3^3}{3}\right) = \frac{37}{63}$$

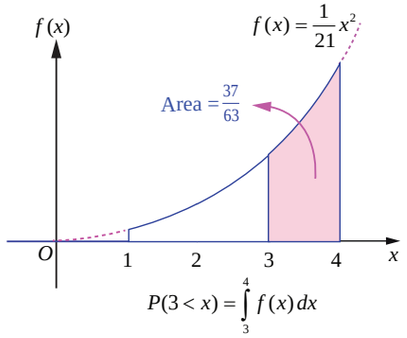


Fig. 11.15

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